METEOROLOGY FORCING

ERT 474/574

Open-Source Hydro Data Analytics

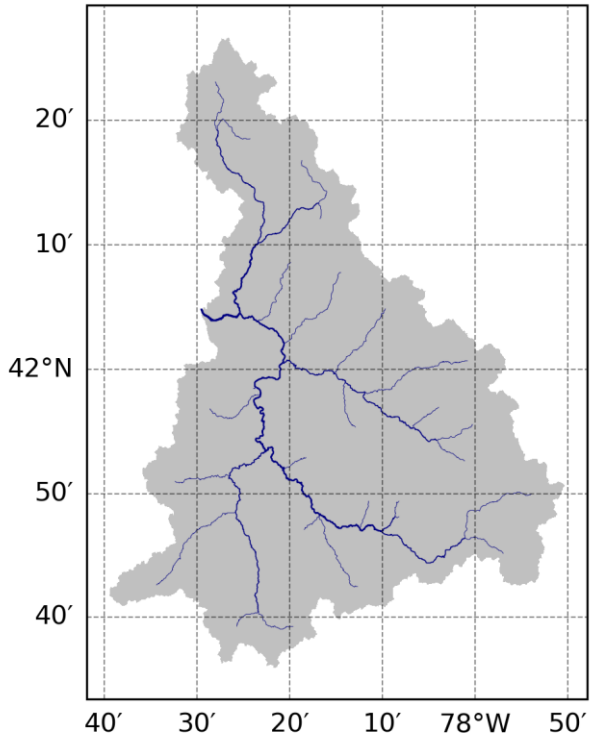
Nov 9th 2026

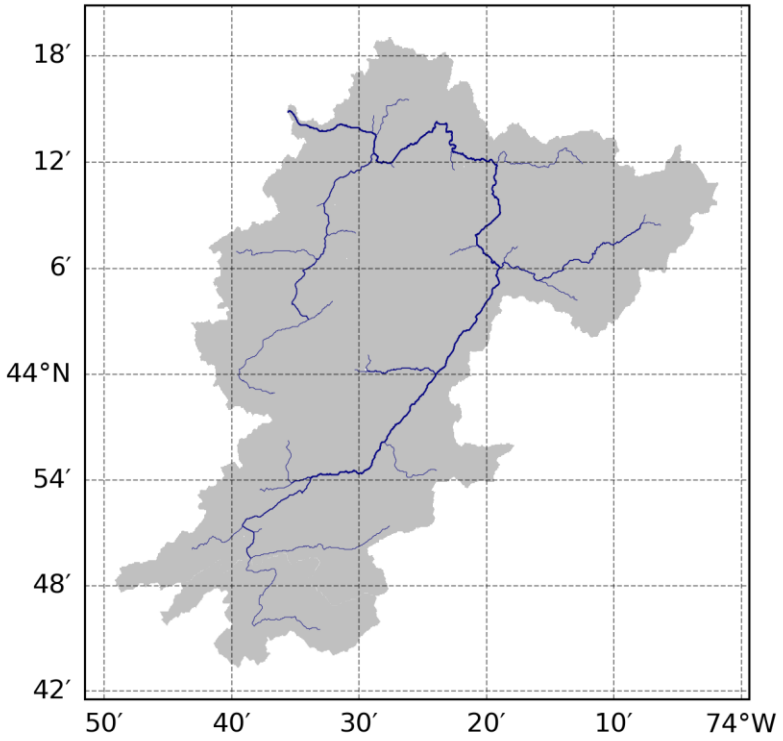
University at Buffalo The State University of New York

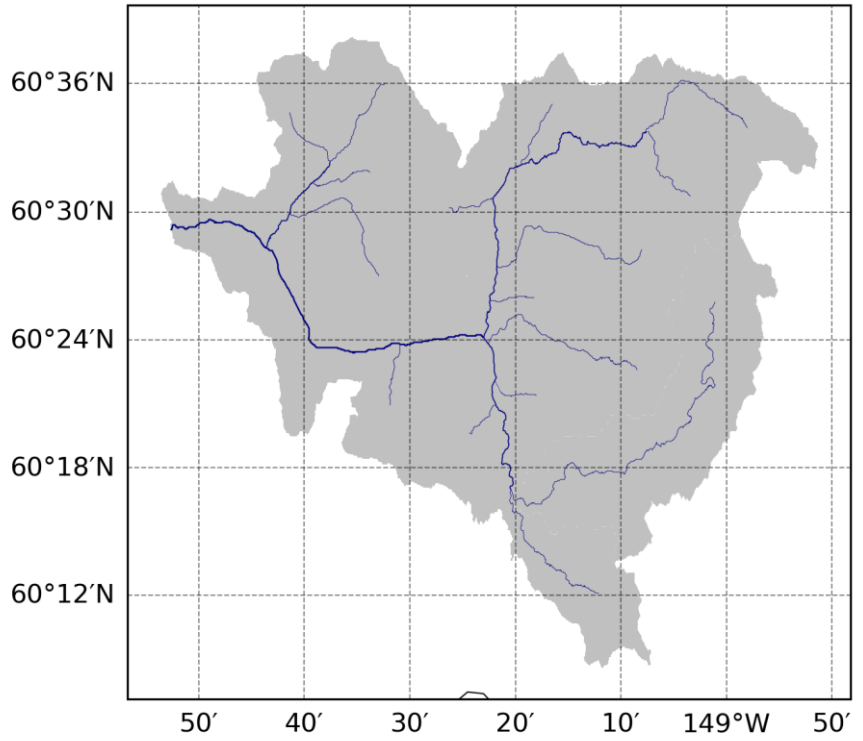


How do we choose basins?

https://waterdata.usgs.gov/explore







Meteorological forcing data

- A set of **weather variables** like temperature, precipitation, humidity, wind speed, air pressure, etc., which are used as input data to drive hydrological models

Meteorological Forcings, Required in all simulations:

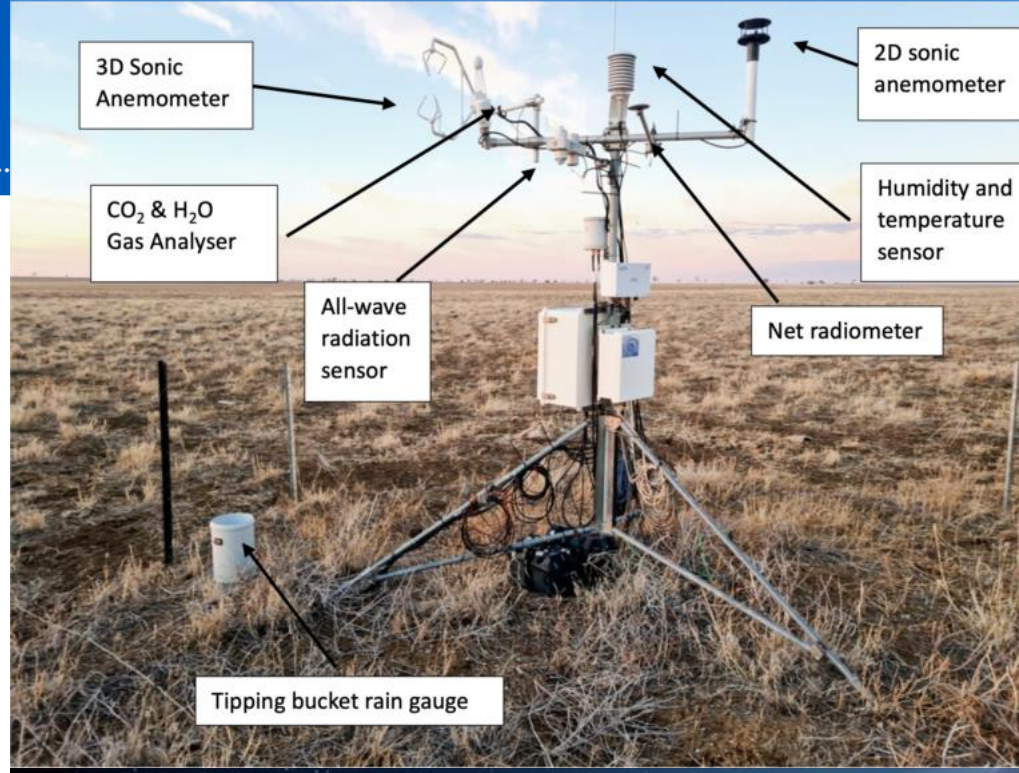
Variable	Description	Units
AIR_TEMP	Average air temperature	C
PREC	Total precipitation (rain and snow)	mm
PRESSURE	Atmospheric pressure	kPa
SWDOWN	Incoming shortwave radiation	W/m^2
LWDOWN	Incoming longwave radiation	W/m^2
VP	Vapor pressure	kPa
WIND	Wind speed	m/s

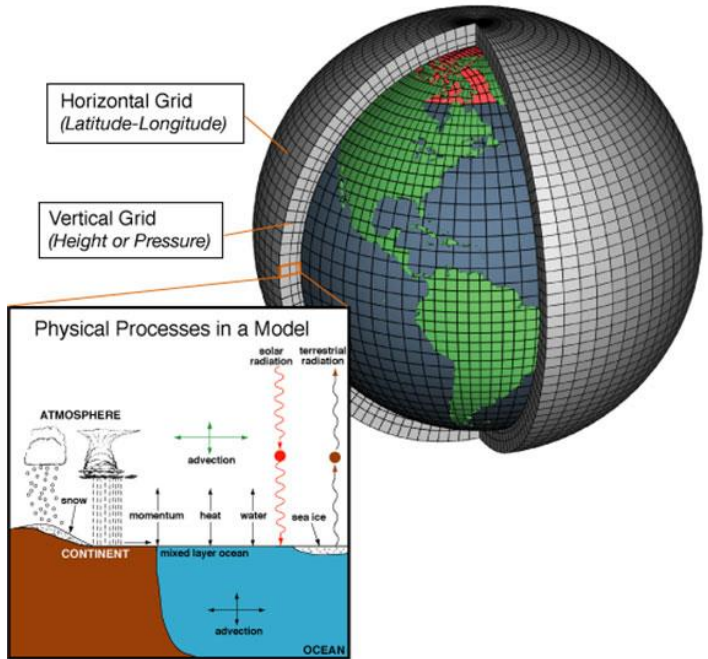
mm/timestep

This value would be different for 3-hourly versus daily timesteps

Source of meteorological data

- Onsite observations
- Remote sensing
- Earth System Models







Ameriflux

- Ameriflux is a network of PI-managed sites measuring ecosystem CO2, water, and energy fluxes in North, Central, and South America.

- It usually provides measurements of all necessary variables for hydrologic modeling
  - Extra quality control might be required

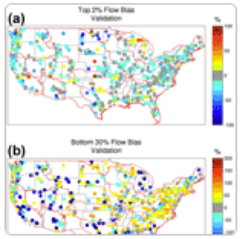
https://ameriflux.lbl.gov/



Large sample watershed-scale dataset for the contiguous U.S.

Research article | **Highlight paper** | 14 Jan 2015

Development of a large-sample watershed-scale hydrometeorological data set for the contiguous USA: data set characteristics and assessment of regional variability in hydrologic model performance



A. J. Newman, M. P. Clark, K. Sampson, A. Wood, L. E. Hay, A. Bock, R. J. Viger, D. Blodgett, L. Brekke, J. R. Arnold, T. Hopson, and Q. Duan

Research article | **Highlight paper** | 20 Oct 2017

The CAMELS data set: catchment attributes and meteorology for large-sample studies



Nans Addor, Andrew J. Newman, Naoki Mizukami, and Martyn P. Clark

Abstract. We present a new data set of attributes for 671 catchments in the contiguous United States (CONUS) minimally impacted by human activities. This complements the daily time series of

Large sample watershed-scale dataset globally

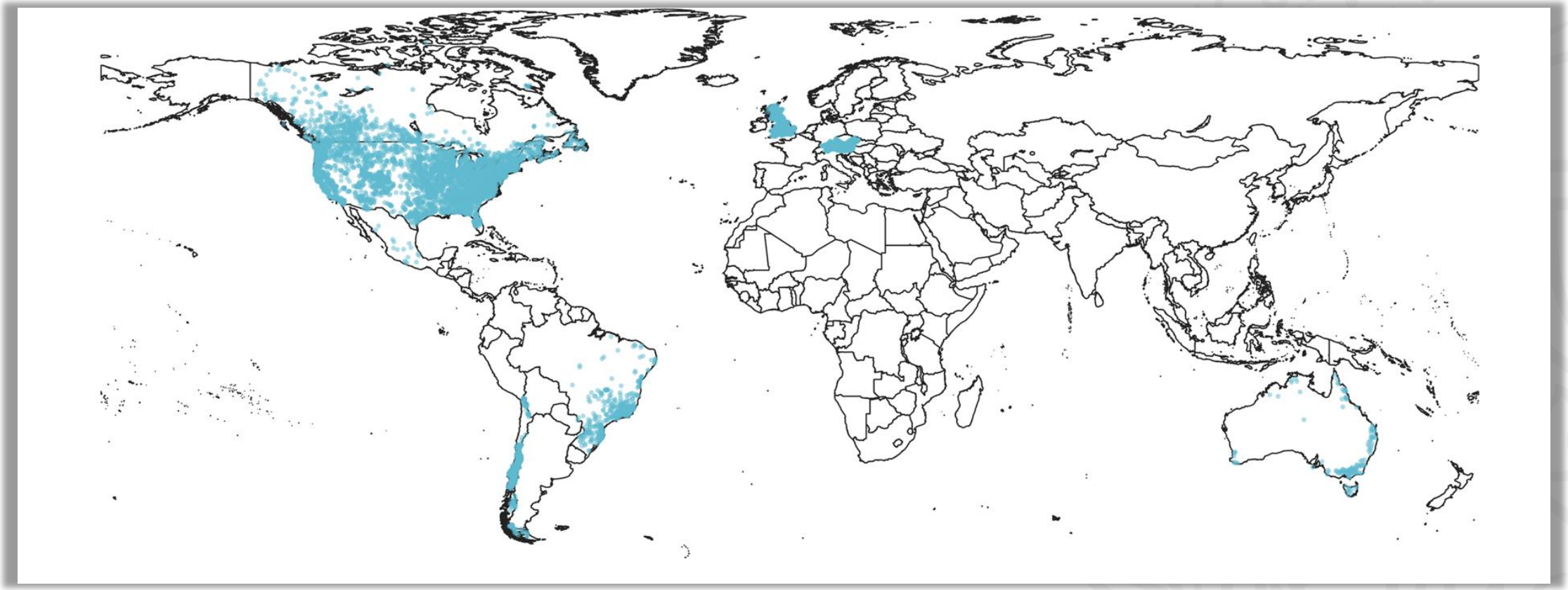
Data Descriptor | Open access | Published: 31 January 2023

Caravan - A global community dataset for large-sample hydrology

Frederik Kratzert, Grey Nearing, Nans Addor, Tyler Erickson, Martin Gauch, Oren Gilon, Lukas Gudmundsson, Avinatan Hassidim, Daniel Klotz, Sella Nevo, Guy Shalev & Yossi Matias

Scientific Data **10**, Article number: 61 (2023) | Cite this article

34k Accesses | **194** Citations | **278** Altmetric | Metrics

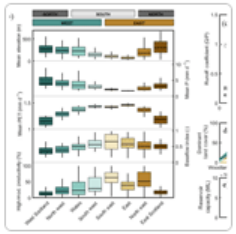


Large sample watershed-scale dataset globally

Data description paper | 12 Oct 2020

CAMELS-GB: hydrometeorological time series and landscape attributes for 671 catchments in Great Britain

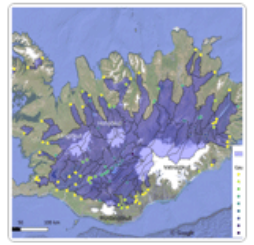
Gemma Coxon, Nans Addor, John P. Bloomfield, Jim Freer, Matt Fry, Jamie Hannaford, Nicholas J. K. Howden, Rosanna Lane, Melinda Lewis, Emma L. Robinson, Thorsten Wagener, and Ross Woods



Data description paper | 13 Jun 2024

LamaH-Ice: LArge-SaMple DAta for Hydrology and Environmental Sciences for Iceland

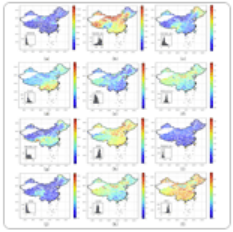
Hordur Bragi Helgason and Bart Nijssen



Data description paper | 03 Dec 2021

CCAM: China Catchment Attributes and Meteorology dataset

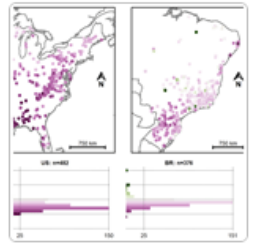
Zhen Hao, Jin Jin, Runliang Xia, Shimin Tian, Wushuang Yang, Qixing Liu, Min Zhu, Tao Ma, Chengran Jing, and Yanning Zhang



Research article | **Highlight paper** | 12 Sep 2024

Large-sample hydrology – a few camels or a whole caravan?

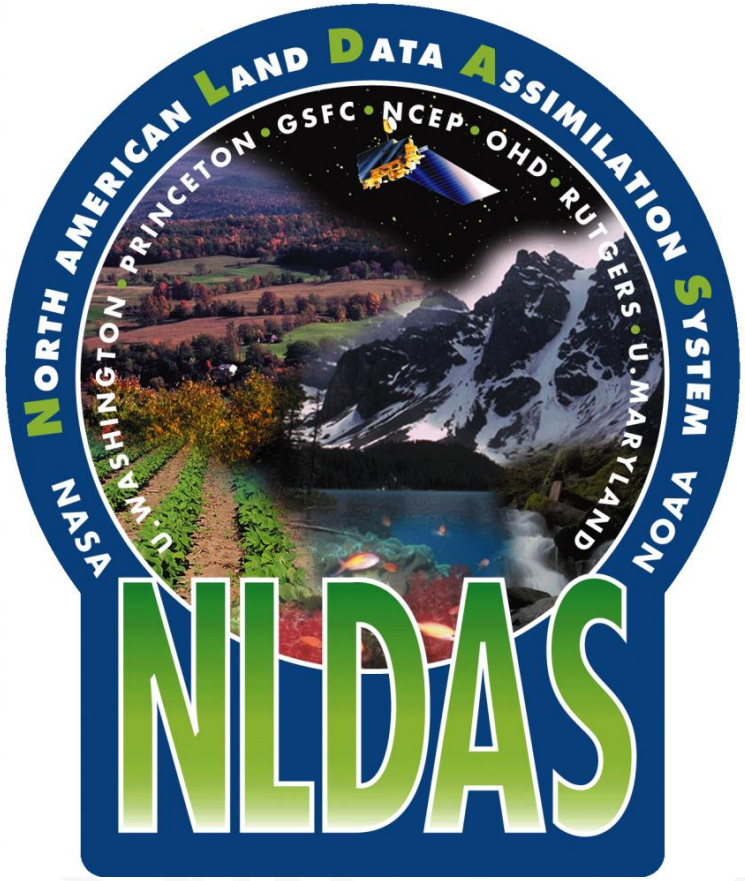
Franziska Clerc-Schwarzenbach, Giovanni Selleri, Mattia Neri, Elena Toth, Ilja van Meerveld, and Jan Seibert



Grid-based meteorological datasets (operational)

- ERA5
  - https://www.ecmwf.int/en/forecasts/dataset/ecmwf-reanalysis-v5
  - ERA5 is the fifth generation ECMWF reanalysis for the global climate and weather for the past 8 decades
- GLDAS/NLDAS
  - https://ldas.gsfc.nasa.gov/nldas
  - North American Land Data Assimilation System (NLDAS) aims to construct quality-controlled, and spatially and temporally consistent, land-surface model (LSM) datasets from the best available observations and reanalyses to support modeling activities.

ECMWF



Grid-based meteorological datasets

- Daymet (https://daymet.ornl.gov/)
  - Daymet provides long-term, continuous, gridded estimates of daily weather and climatology variables by interpolating and extrapolating ground-based observations through statistical modeling techniques.
- GMET (https://ncar.github.io/hydrology/models/GMET)
  - Gridded Meteorological Ensemble Tool (GMET) allows for quantification of uncertainty for station-based gridded precipitation and temperature datasets
- gridMET (https://www.climatologylab.org/gridmet.html)
  - gridMET is a dataset of daily surface meteorological data covering the contiguous US from 1979-yesterday.

Compared to ERA5 and NLDAS, what are the major differences for these datasets?

Grid-based meteorological datasets

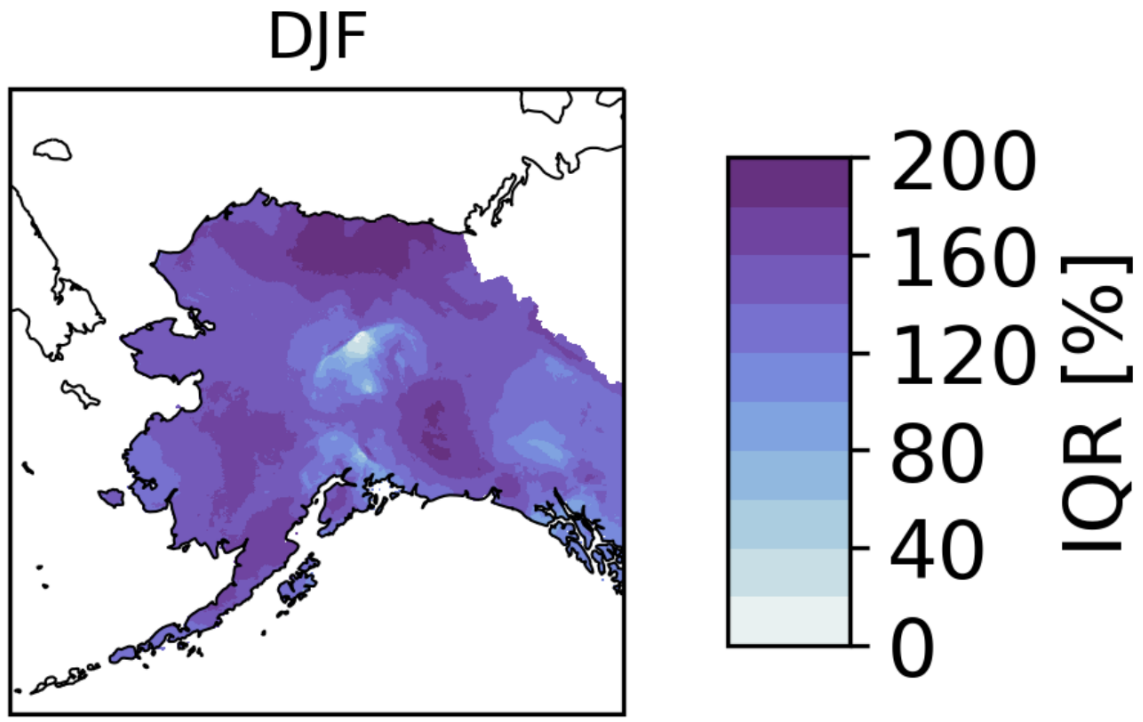
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High resolution (1km)

Uncertainty Quantification

Semi-operational (Downscaled NLDAS)

Biases exist in the gridded meteorological forcings



Example: Relative Interquartile Range for GMET data in Alaska

$$IQR = \frac{P_{75\%} - P_{25\%}}{(P_{75\%} + P_{25\%})/2}$$

Biases in the met forcings are major sources for biases in the hydrologic modeling simulations!

Example: Steps to process meteorological forcings from NLDAS

- Step 1: Download the datasets
- Step 2: Concatenate the hourly files into the targeted format
- Step 3: Convert the units of the variables to the units required by the VIC model
- Step 4: If necessary, regridding the met forcing data to the targeted domain grid

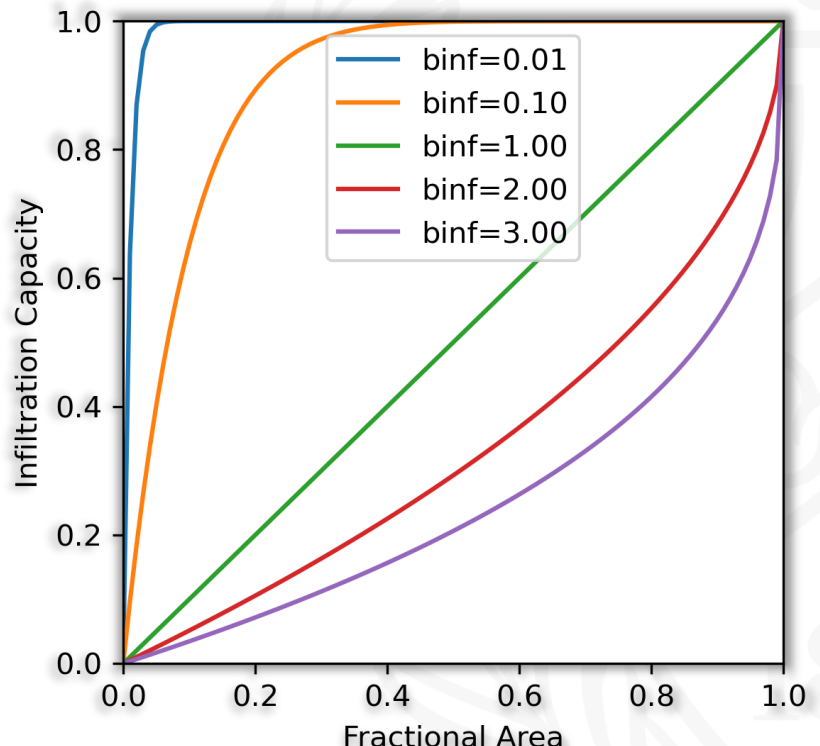
Live Demo
https://ldas.gsfc.nasa.gov/nldas

General rules when preparing meteorological forcings…

When the spatial resolutions of land surface/parameter datasets and meteorological forcing are different, it is recommended to **regrid the met forcing**.

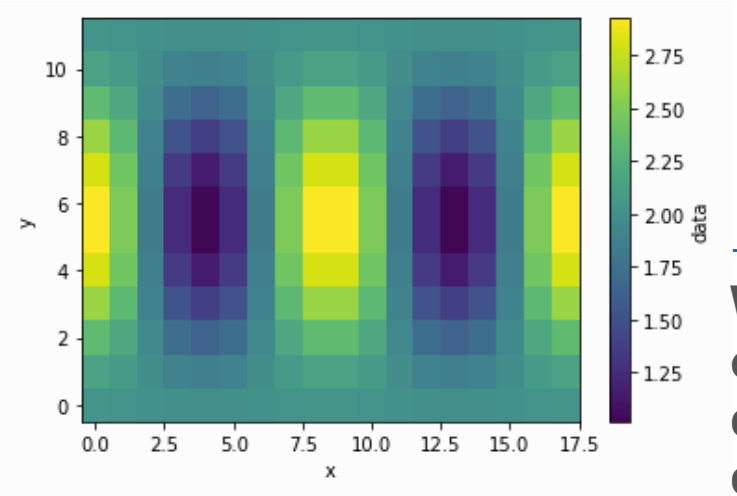
Question: Why don't we regrid land surface datasets?

20% clay 80% silt B_{inf}=0.1	60% clay 40% silt B_{inf}=1



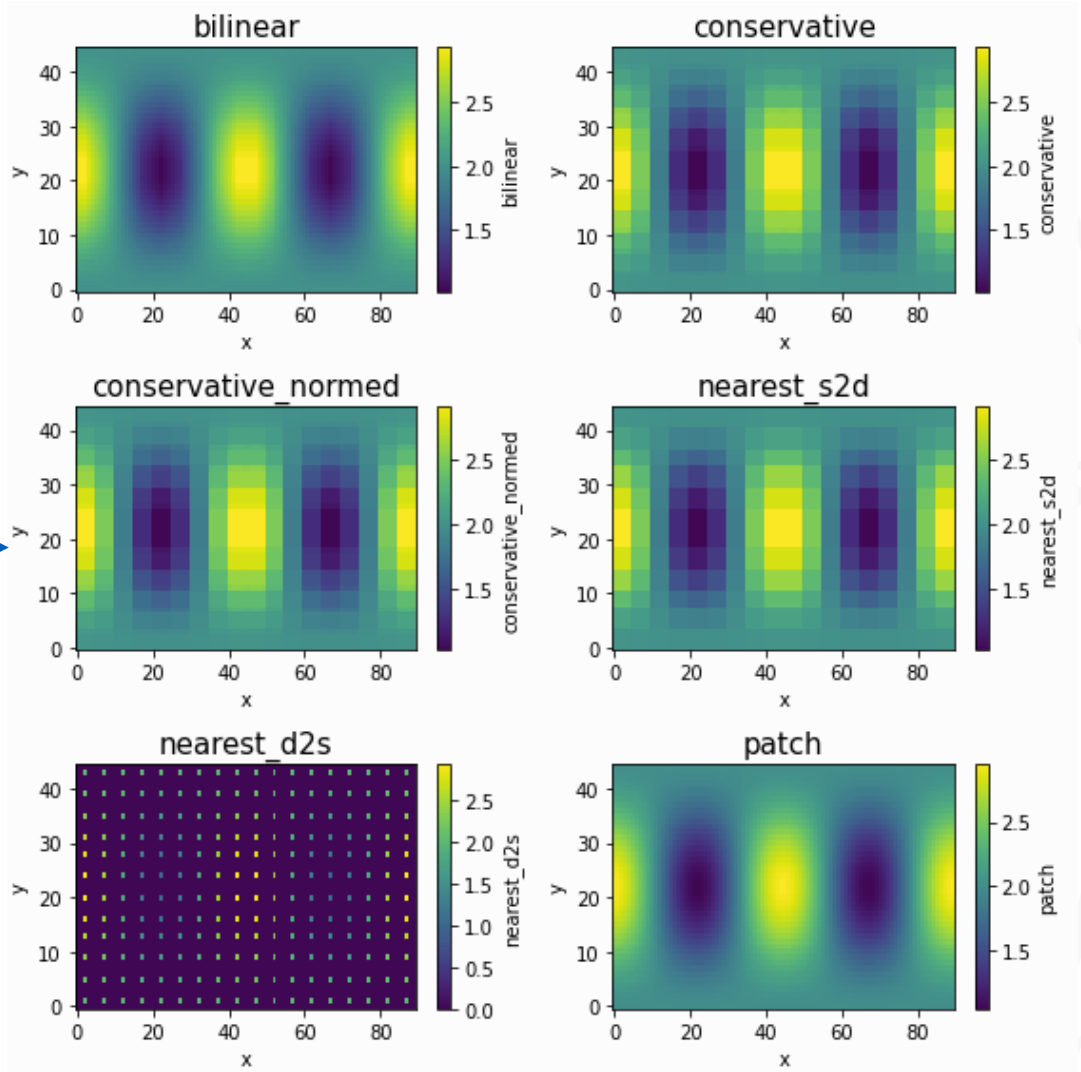
The arithmetic average works when calculating soil composition but it won't work when calculating non-linear parameters!

Regridding



Regrid

When the spatial resolution of meteorological forcing is different than the model domain

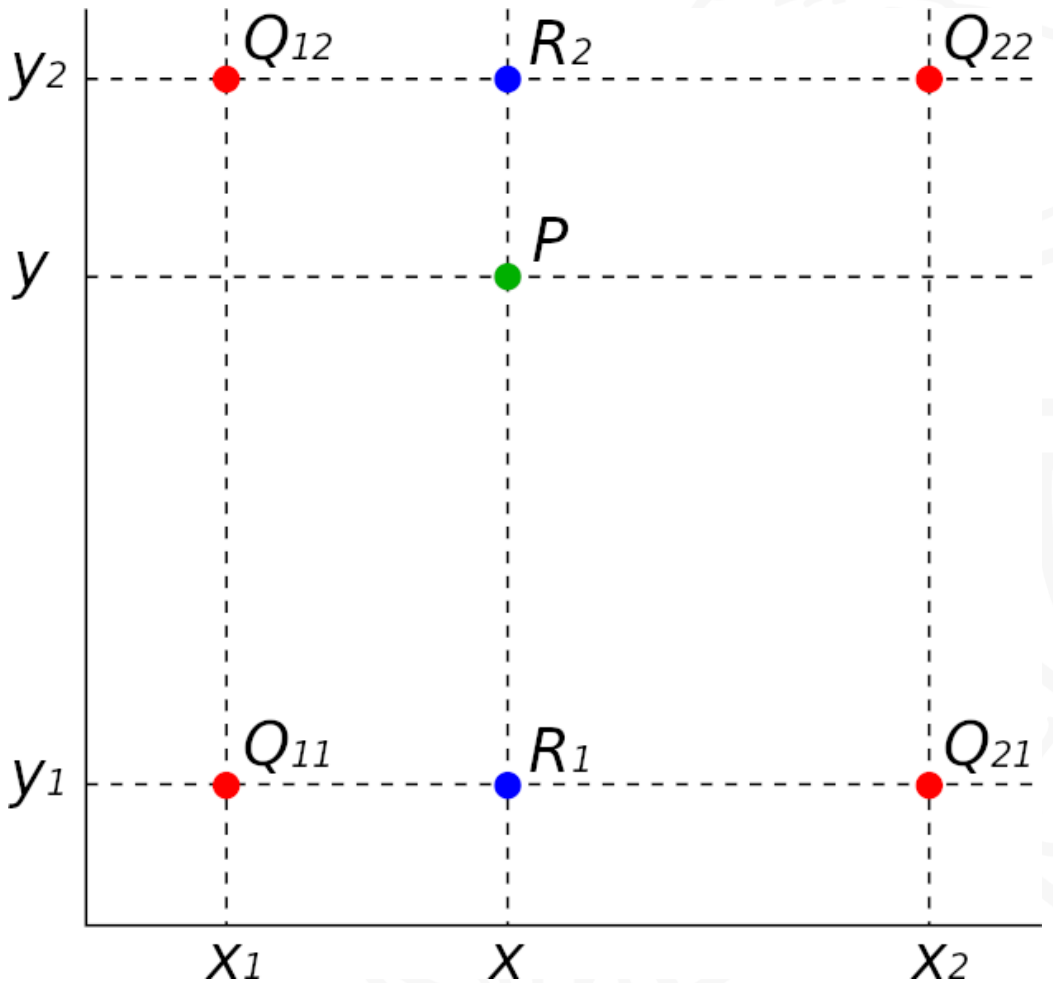


https://xesmf.readthedocs.io/en/stable/notebooks/Compare_algorithms.html#Comparison-of-six-regridding-algorithms

Bilinear

How does bilinear (2-D) interpolation work?

Let's assume that we have defined a set of data coordinates (x_k, y_k), where k = 1, 2. These coordinates define the position of the points Q_{11}, Q_{21}, Q_{12}, and Q_{22}. For any given x and y coordinates, which are located between the x_k and y_k points, by applying the bilinear interpolation technique, we can find the P point (defined by x and y).



Questions?

